MAST20026 Real Analysis Semester 2 2025

Assignment 4

Due: 25 September, 11:59pm

M= a mark associated with applying an appropriate method

J= a mark associated with using correct justification

A= a mark associated with a correct answer or intermediate step

N= a mark associated with good notation or writing

Please see the assignment instructions on the Assignment 4 page on the LMS, where you accessed this file. To ease submission, please prepare assignment solutions for each question on its own page. This document is 2 pages in length.

PART A (*2 marks* total)

Go to Corrine Yap's webpage on her play *Uniform Convergence* linked from the Assignment 4 page on the LMS. There is a link to a recording of her performance of the play, as well as a blogpost on where the ideas for the play originated. Watch the performance and/or read the blog post, and then write a short reflective response (around half a page) that considers the following questions.

- Was there anything about the performance and/or the blog that you found surprising?
- What (if anything) resonated with you about the performance and/or blog? This can be a sensitive topic for some people, so share only what you feel comfortable doing. You won't be penalised for not responding to this prompt.
- Do you think the performance was art, mathematics, or both? Is there a place for art in mathematics, and is there a place for mathematics in the arts?

1 mark for demonstrating thoughtful engagement with the task.
1 mark for clearly communicated response (the response is well written, and conveys thoughts concisely).

PART B (*20 marks* total)

1. **Cauchy sequences.** *(6 marks)* Let (f_n) be a sequence such that

$$f_{n+1} = f_n + \frac{1}{n!}$$

for all $n \in \mathbb{N}^+$. Prove that (f_n) is Cauchy.
If needed, you can use that $n! > 2^n$ for $n \geq 4$ without proof.

Solution.

1M: Demonstrated understanding of the definition of being Cauchy (eg. from the structure of the proof)

First observe that $(\forall n \in \mathbb{N}^+)\ |f_{n+1} - f_n| = \frac{1}{n!}$.
Let $\epsilon > 0$.
Define $M = \max\{4, \lceil \log_2(\frac{1}{\epsilon}) + 1 \rceil\}$. 1A: Any valid M that works for the proof of being Cauchy
Assume $m, n > M$. Without loss of generality, assume $m \geq n$.
Then

1J: A mostly correct sequence of inequalities

$$
\begin{aligned}
&|f_m - f_n| \\
&= |f_m - f_{m-1} + f_{m-1} - f_{m-2} + \cdots - f_{n+1} + f_{n+1} - f_n| \\
&\leq |f_m - f_{m-1}| + |f_{m-1} - f_m| + \cdots + |f_{n+1} - f_n| && \text{triangle inequality} \\
&\leq \frac{1}{(m-1)!} + \cdots + \frac{1}{n!} && |f_{k+1} - f_k| = \frac{1}{k!} \\
&\leq \frac{1}{2^{m-1}} + \cdots + \frac{1}{2^n} && n > M \geq 4, \text{ and } n! > 2^n \text{ for } n \geq 4 \\
&\leq \frac{1}{2^n}(1 + \frac{1}{2} + \cdots + \frac{1}{2^{m-1-n}}) \\
&= \frac{1}{2^n} \frac{1 - \frac{1}{2^{m-n}}}{1 - \frac{1}{2}} && \text{geometric sum} \\
&= \frac{1}{2^{n-1}} \left(1 - \frac{1}{2^{m-n}}\right) \\
&\leq \frac{1}{2^{n-1}} \\
&< \epsilon && n > M \geq \lceil \log_2(\frac{1}{\epsilon}) + 1 \rceil
\end{aligned}
$$

1J: explicit justification included when using assumptions on n, m, and properties of the sequence provided in the question

1A: A complete and correct sequence of inequalities showing that |f_m - f_n| < epsilon

Therefore (a_n) is a Cauchy sequence.

Remark: Another approach to this problem is to show that the sequence is contractive, since we showed in lectures that contractive sequences are Cauchy.

1N: The proof is correctly structured and easy to follow.

2. $\epsilon - \delta$ **definition**. *(6 marks)* Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$, where

$$f(x) = x^3 + x^2 + x + 1 + \frac{1}{x}.$$

Use the $\epsilon - \delta$ definition of limits to show that

$$\lim_{x \to 1} f(x) = 5.$$

This calculation has the potential to be a bit messy. Draft your response first, and aim to produce a polished proof for submission.

Solution. 1M: Demonstrated understanding of the epsilon-delta definition of continuity

Let $\epsilon > 0$.

Let $\delta = \min(\{\frac{1}{2}, \frac{4}{41\epsilon}\})$ 1A: Any valid delta that works

Assume $|x - 1| < \delta$.

$$\begin{aligned}
|f(x) - 5| &= |x^3 + x^2 + x + 1 + \frac{1}{x} - 5| \\
&= |x^3 + x^2 + x - 4 + \frac{1}{x}| \\
&= |x - 1||x^2 + 2x + 3 - \frac{1}{x}| \\
&\leq |x - 1|(|x|^2 + 2|x| + 3 + \frac{1}{|x|}) \quad \text{(triangle inequality)} \\
&\leq |x - 1|(\frac{9}{4} + 3 + 3 + 2) \quad (\text{ since } |x - 1| < \frac{1}{2} \implies \frac{1}{2} < |x| < \frac{3}{2}\ ) \\
&= \frac{41}{4}|x - 1| \\
&< \epsilon \quad (\text{ since } |x - 1| < \delta \leq \frac{4}{41\epsilon}\ )
\end{aligned}$$

1J: A mostly correct sequence of inequalities

1A: A complete and correct sequence of inequalities showing that |f(x)-5| < epsilon

1J: explicit justification included when using assumptions |x-1|

1N: The proof is correctly structured and easy to follow.

3. **Limit points**. *(8 marks)* For a subset $E \subseteq \mathbb{R}$, define $L(E)$ to be the set of limit points of E.

   (a) Give an example of a subset $A \subseteq \mathbb{R}$ such that $L(A) \subset A$ and $L(A) \neq A$, or explain why no such example exists.

   (b) Give an example of a subset $B \subseteq \mathbb{R}$ such that $B \subset L(B)$ and $L(B) \neq B$, or explain why no such example exists.

   (c) Give an example of a finite subset $C \subseteq \mathbb{R}$ such that $L(C) = \{1\}$, or explain why no such example exists.

   (d) Give an example of a two subsets $X, Y \subseteq \mathbb{R}$ such that $X \cap Y \neq \emptyset$, and $L(Y) \neq \emptyset$, and $L(X) \neq \emptyset$ and $L(X \cap Y) = \emptyset$, or explain why no such example exists.

Briefly justify your responses.

Solution.

1A: A correct example
1J: A brief and correct reason is provided

(a) Let $A = [0,1] \cup \{2\}$.
Note that 2 is not a limit point of A since there are no points of A in $(1.5, 2.5)$.
So $L(A) = [0,1]$.
So $L(A) \subset A$ but $L(A) \neq A$.

1A: A correct example
1J: A brief and correct reason is provided

(b) Let $B = (0,1)$.
From class (or otherwise) $L(B) = [0,1]$. So $B \subset L(B)$ but $L(B) \neq B$.

1A: No example exists
1J: A brief and correct reason is provided

(c) No such example exists.
A sketch of a proof / brief reason is:
Let $C = \{x_1, \ldots, x_n\}$.
Let $\delta = min\{|x_1 - 1|, \ldots, |x_n - 1|\}$. (At this step, take out x_j if $x_j = 1$ so $\delta > 0$).
Consider $\frac{\delta}{2} > 0$.
Then there are no other points of C in $(1 - \frac{\delta}{2}, 1 + \frac{\delta}{2})$.

1A: A correct example
1J: A brief and correct reason is provided

(d) Let $X = [0,1]$ and $Y = [1,2]$.
Then $L(X) = X \neq \emptyset$ and $L(Y) = Y \neq \emptyset$.
However, $X \cap Y = \{1\} \neq \emptyset$.
Since $X \cap Y$ has one element, $L(X \cap Y) = \emptyset$.